

Part II V2 in Lamport notation

Same-noise interaction comparison, edge correction, and shifted repair

Fresh reconstruction of `part_ii/v2_interaction_comparison.tex`

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This reconstruction corrects the false statement that four edge modes cannot conserve momentum and replaces the incomplete proof of the shifted comparison by an exact fundamental-band/alias partition. The repaired estimates have the rates claimed in V2. The original V2 file is not valid without these additions.

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1 Definitions

Fix $t \geq t_0 > 0$, $L, m > 0$, and assume $t_0 L \geq 1$. Put

$$D_N = \frac{2\pi}{t} \mathbb{Z} \times \Gamma_N, \quad D_\infty = \frac{2\pi}{t} \mathbb{Z} \times \frac{\pi}{L} \mathbb{Z}, \quad R_N = \frac{\pi}{\varepsilon_N}.$$

For $\kappa = (k_0, k)$ define

$$c_N(\kappa) = \frac{1}{k_0^2 + \gamma_N(k)^2}, \quad c_\infty(\kappa) = \frac{1}{k_0^2 + m^2 + k^2}, \quad c_\times(\kappa) = \sqrt{c_N(\kappa)c_\infty(\kappa)}. \quad (0.1)$$

Let $k_e = -R_N$. On one real Gaussian Fourier family $G(-\kappa) = \overline{G(\kappa)}$ set

$$G_N(k_0, k) = G(k_0, k) \quad (k \neq k_e), \quad G_N(k_0, k_e) = 2^{-1/2}[G(k_0, k_e) + G(k_0, -k_e)]. \quad (0.2)$$

Then

$$\eta_N(z) = (2Lt)^{-1/2} \sum_{D_N} G_N(\kappa) c_N(\kappa)^{1/2} e^{i\kappa z}, \quad \eta(z) = (2Lt)^{-1/2} \sum_{D_\infty} G(\kappa) c_\infty(\kappa)^{1/2} e^{i\kappa z}. \quad (0.3)$$

2 Structured proof

$\langle 1 \rangle$ 1 Determine the laws and the edge covariance exactly.

$\langle 2 \rangle$ 1 η_N is real.

Because: Away from the edge, the terms at κ and $-\kappa$ are conjugate. On the edge, the lattice characters at $\pm R_N$ agree, and

$$G_N(-k_0, k_e) = 2^{-1/2}[G(-k_0, k_e) + G(-k_0, -k_e)] = \overline{G_N(k_0, k_e)}.$$

$\langle 2 \rangle 2$ The G_N are orthonormal on D_N and have the real-field pairing.

Because: Off the edge this is the covariance of G . On the edge,

$$\mathbb{E}[G_N(k_0, k_e)\overline{G_N(k'_0, k_e)}] = \frac{1}{2}(1+1)\delta_{k_0 k'_0},$$

while cross terms between $+R_N$ and $-R_N$ noises vanish. Expanding the two two-term averages without conjugation leaves exactly the pairings with $k'_0 = -k_0$, and therefore $\mathbb{E}[G_N(k_0, k_e)G_N(k'_0, k_e)] = \delta_{k_0, -k'_0}$.

$\langle 2 \rangle 3$ Therefore

$$\mathbb{E}[\eta_N(s, x)\eta_N(s', y)] = \frac{1}{2L} \sum_{k \in \Gamma_N} e^{ik(x-y)} c_{\gamma_N(k)}(s - s'), \quad (1.1)$$

$$c_\omega(\tau) = \frac{\cosh(\omega(t/2 - |\tau|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}.$$

Because: Fourier pairing gives the sum, and

$$\frac{1}{t} \sum_{k_0 \in (2\pi/t)\mathbb{Z}} \frac{e^{ik_0\tau}}{k_0^2 + \omega^2} = c_\omega(\tau).$$

Both sides solve $(-\partial_\tau^2 + \omega^2)c = \delta_{\mathbb{T}_t}$; the derivative jump at zero is -1 , which fixes the Fourier coefficients and normalization.

$\langle 2 \rangle 4$ The mixed covariance is

$$\mathbb{E}[\eta_N(z)\eta(z')] = C_\times(z - z') + \mathcal{R}_N(z, z'), \quad (1.2)$$

where $C_\times$ is the D_N Fourier sum with symbol $c_\times$ and

$$\mathcal{R}_N = (2Lt)^{-1} \sum_{k_0} c_\times(k_0, k_e) e^{ik_0(s-s')} \left[2^{-1/2}(e^{ik_e(x-y)} + e^{ik_e(x+y)}) - e^{ik_e(x-y)} \right]. \quad (1.3)$$

Because: Only (0.2) differs from the common-noise pairing. Pairing its two summands with the continuum noises at $\pm R_N$ gives the two characters in (1.3); subtracting the surrogate D_N term gives the bracket.

$\langle 2 \rangle 5$ For $\varepsilon_N \leq 1/2$,

$$\|\mathcal{R}_N\|_\infty \leq \frac{3 \coth(2t_0)}{8L} \varepsilon_N. \quad (1.4)$$

Because: The bracket is at most 3, $c_\times(k_0, k_e) \leq (k_0^2 + \gamma_e^2)^{-1}$, $\gamma_e \geq 2/\varepsilon_N$, and

$$\sum_{k_0} \frac{1}{k_0^2 + \gamma_e^2} = \frac{t}{2\gamma_e} \coth(t\gamma_e/2).$$

$\langle 2 \rangle 6$ Equivalently, the true mixed kernel is the Fourier sum over the symmetric band $\tilde{\Gamma}_N = \Gamma_N \cup \{+R_N\}$ with edge weights $\chi(\pm R_N) = 2^{-1/2}$ and $\chi = 1$ otherwise.

Because: Combine the two edge characters in (1.3) with the original edge term. On lattice points $e^{iR_N x} = e^{-iR_N x}$, so both continuum partners represent the same lattice degree of freedom.

⟨2⟩ 7 This proves ⟨1⟩1. □

⟨1⟩ 2 Establish Wick recombination and existence of the continuum integrated chaoses.

⟨2⟩ 1 For a centered Gaussian X of variance v and any c ,

$$:X^4:_c = :X^4:_v + 6(v - c):X^2:_v + 3(v - c)^2. \quad (2.1)$$

Because: Expand both sides using $:X^4:_a = X^4 - 6aX^2 + 3a^2$ and $:X^2:_a = X^2 - a$; coefficients of $X^4, X^2, 1$ agree.

⟨2⟩ 2 With lattice thermal variance $v_N(t)$ and vacuum Wick constant $c_{N,L}$,

$$d_N(t) := v_N(t) - c_{N,L} = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{1}{\gamma_N(k)(e^{t\gamma_N(k)} - 1)}. \quad (2.2)$$

Moreover $0 \leq d_N \leq \bar{d} < \infty$ and $|d_N - d_\infty| \leq C\varepsilon_N^2$ uniformly for $t \geq t_0$.

Because: $\coth(y/2) - 1 = 2/(e^y - 1)$. On the band, $\gamma_N(k) \geq \max(m, 2|k|/\pi)$; the positive majorant decays exponentially. For common modes apply the mean value theorem to $f_t(y) = [y(e^{ty} - 1)]^{-1}$ and $0 \leq \gamma(k) - \gamma_N(k) \leq \varepsilon_N^2 k^4/(24m)$. The derivative is bounded by $C(m, t_0)(1 + t)e^{-t\gamma_-(k)/2}$, whose k^4 -weighted sum is uniform in $t \geq t_0$. Missing spatial modes contribute $O(e^{-2t_0/\varepsilon_N})$.

⟨2⟩ 3 For jointly centered Gaussian X, Y ,

$$\mathbb{E}[:X^j:_v :Y^\ell:_v] = \delta_{j\ell} j! \mathbb{E}[XY]^j. \quad (2.3)$$

Because: Multiply the generating functions $e^{\alpha X - \alpha^2 v_X/2}$ and $e^{\beta Y - \beta^2 v_Y/2}$; their expectation is $e^{\alpha\beta \mathbb{E}[XY]}$. Compare the absolutely convergent power-series coefficients.

⟨2⟩ 4 Let $Y_j^{(K)} = \int_{\mathbb{T}_t \times \mathbb{T}_L} : \eta_K^j :$ for $j = 2, 4$. Then

$$\mathbb{E}[Y_j^{(K)} Y_j^{(K')}] = j!(2Lt)^{-(j-2)} \sum_{\substack{\kappa_i \in D_\infty^{\leq K \wedge K'} \\ \sum_i \kappa_i = 0}} \prod_i c_\infty(\kappa_i). \quad (2.4)$$

Because: Apply (2.3) pointwise, expand the covariance Fourier series, and integrate over the two spacetime variables. Each integration supplies $2Lt$ if the total momentum is zero and zero otherwise.

⟨2⟩ 5 The sums in (2.4) increase with K and are bounded; hence $Y_j^{(K)}$ is Cauchy in L^2 and defines Y_j .

Because: Put $c_{\text{env}}(\kappa) = (\pi^2/4)(m^2 + |\kappa|^2)^{-1}$. Step ⟨1⟩3 proves the convolution bounds that make $\sum c_{\text{env}}(\kappa)(c_{\text{env}}^*)(-\kappa)$ and $\sum c_{\text{env}}^2$ finite. Positivity gives monotonicity; the polarization identity gives the Cauchy norm.

⟨2⟩ 6 Define

$$\begin{aligned} \mathcal{U}_N &= \lambda[X_4 + 6d_N X_2 + 3d_N^2(2Lt)], \\ \mathcal{U} &= \lambda[Y_4 + 6d_\infty Y_2 + 3d_\infty^2(2Lt)]. \end{aligned} \quad (2.5)$$

Then $\mathcal{U}_N$ is exactly the lattice-vacuum Wick quartic integrated over spacetime.

Because: Apply (2.1) at every spacetime point with $v = v_N$ and $c = c_{N,L}$, then sum and integrate.

⟨2⟩ 7 This proves ⟨1⟩2. □

⟨1⟩ 3 Prove the covariance envelope and convolution estimates.

⟨2⟩ 1 $c_N, c_\times, c_\infty \leq c_{\text{env}}$.

Because: On $|\varepsilon_N k| \leq \pi$, $\sin(|\varepsilon_N k|/2) \geq |\varepsilon_N k|/\pi$, so $k_0^2 + \gamma_N(k)^2 \geq (4/\pi^2)(m^2 + k_0^2 + k^2)$. By its definition, $c_\infty = (m^2 + k_0^2 + k^2)^{-1} \leq c_{\text{env}}$; hence $c_\times = (c_N c_\infty)^{1/2} \leq c_{\text{env}}$ as well.

⟨2⟩ 2 For $R \geq 0$, the anisotropic grid satisfies

$$\sum_{|\kappa| \leq R} c_{\text{env}}(\kappa) \leq C(m, t_0) tL [1 + \log(2 + R)], \quad (3.1)$$

$$\sum_{|\kappa| > R} \frac{1 + \log(2 + |\kappa|)}{(m^2 + |\kappa|^2)^2} \leq C(m, t_0) tL \frac{1 + \log(2 + R)}{m^2 + R^2}. \quad (3.2)$$

Because: Cover each grid point by its rectangle of side lengths $2\pi/t$ and π/L . Outside a fixed disk the radial functions vary by a bounded factor on one rectangle and the sum is bounded by the grid density $tL/(2\pi^2)$ times the radial integral. Inside the disk there are finitely many rectangles; the isolated maximum is absorbed because $t \geq t_0$ and $tL \geq 1$. The radial integrals are $\int_0^R r(m^2 + r^2)^{-1} dr = O(1 + \log(2 + R))$ and $\int_R^\infty r(1 + \log(2 + r))(m^2 + r^2)^{-2} dr = O((1 + \log(2 + R))/(m^2 + R^2))$.

⟨2⟩ 3 For $F_2 = c_{\text{env}} * c_{\text{env}}$,

$$F_2(\chi) \leq C(m, t_0) tL \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}. \quad (3.3)$$

Because: Split the summation into $|\kappa| \leq |\chi|/2$, $|\chi - \kappa| \leq |\chi|/2$, and the complement. On the first region the second factor is $O((m^2 + |\chi|^2)^{-1})$ and (3.1) sums the first; the second region is symmetric. On the complement both arguments exceed $|\chi|/2$; replace the product by twice the square of the factor with smaller argument and apply (3.2) without its logarithm.

⟨2⟩ 4 For $F_3 = c_{\text{env}} * F_2$,

$$F_3(\chi) \leq C(m, t_0) (tL)^2 \frac{[1 + \log(2 + |\chi|)]^2}{m^2 + |\chi|^2}. \quad (3.4)$$

Because: Use the same three regions. In the first, sum F_2 up to radius $|\chi|/2$; (3.3) and a radial integral produce two logarithms. In the second, bound F_2 at radius $|\chi|/2$ and sum c_{env} by (3.1). In the complement, the product is bounded by $CtL(1 + \log(2 + |u|))(m^2 + |u|^2)^{-2}$ at the smaller argument; apply (3.2).

⟨2⟩ 5 The dispersion-symbol difference obeys

$$|c_N - c_\infty|(\kappa) \leq \frac{\varepsilon_N^2 k^4}{12} c_N(\kappa) c_\infty(\kappa), \quad |c_\times - c_\infty| \leq |c_N - c_\infty|. \quad (3.5)$$

Because: Subtract the two resolvents and use $0 \leq k^2 - 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2) \leq \varepsilon_N^2 k^4/12$. For the geometric mean, $|\sqrt{a} - \sqrt{b}| = |a - b|/(\sqrt{a} + \sqrt{b})$.

⟨2⟩ 6 This proves ⟨1⟩3. □

⟨1⟩ 4 Prove the unshifted L^2 interaction comparison, including every edge sector.

⟨2⟩ 1 Wick–Isserlis and Fourier orthogonality give

$$\mathbb{E}[X_4^2] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in D_N \\ \sum k_{0,i}=0, \sum \kappa_i \equiv 0 \pmod{2R_N}}} \prod_i c_N(\kappa_i), \quad (4.1)$$

$$\mathbb{E}[Y_4^2] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in D_\infty \\ \sum \kappa_i=0}} \prod_i c_\infty(\kappa_i), \quad (4.2)$$

$$\mathbb{E}[X_4 Y_4] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in \tilde{D}_N \\ \sum \kappa_i=0}} \prod_i \chi(k_i) c_\times(\kappa_i). \quad (4.3)$$

Because: The lattice spatial sum enforces conservation modulo $2R_N$; the continuum spatial integral enforces exact conservation. Step ⟨1⟩1⟨2⟩6 gives the symmetric band and weights in (4.3).

⟨2⟩ 2 The four-edge sector omitted in V2 is

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_i f(k_{0,i}), \quad f(k_0) = c_\times(k_0, R_N). \quad (4.4)$$

Because: Exact spatial conservation with four edge inputs requires two $+R_N$ and two $-R_N$. There are $\binom{4}{2} = 6$ orders and their weight is $(2^{-1/2})^4 = 1/4$, hence $6/4 = 3/2$. Configurations with exactly three edge inputs are impossible, but configurations with four are not.

⟨2⟩ 3 The correction satisfies

$$0 \leq \Delta_{4e} \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5. \quad (4.5)$$

Because: Let $A = \sum f$ and $B = \sum f^2$. From $f \leq (k_0^2 + \gamma_e^2)^{-1}$ and $\gamma_e \geq 2/\varepsilon_N$,

$$A \leq \frac{t\varepsilon_N}{4} \coth(2t_0), \quad B \leq \gamma_e^{-2} A \leq \frac{t\varepsilon_N^3}{16} \coth(2t_0).$$

Furthermore $\sum \sum_{k_{0,i}=0} \prod f = \|f * f\|_2^2 \leq \|f * f\|_\infty \|f * f\|_1 \leq BA^2$. Insert this into (4.4).

⟨2⟩ 4 The remaining one-edge and two-edge differences between the symmetric and half-open descriptions are $O(t\varepsilon_N^3 \log_N^2)$.

Because: For one edge, sum its symbol to obtain $O(t\varepsilon_N)$ and bound the remaining three-fold constrained sum at spatial momentum R_N by (3.4), namely $O((tL)^2 \varepsilon_N^2 \log_N^2)$; the prefactor gives the claimed order. For two edges, put the temporal edge sequence in ℓ^2 and use $\|f * f\|_\infty \leq \sum f^2 = O(t\varepsilon_N^3)$; the remaining two-fold sum is bounded by (3.3). There are finitely many placements. Step ⟨2⟩3 covers four edges and is smaller.

⟨2⟩ 5 The lattice alias contribution in (4.1) obeys

$$Al_N \leq C(m, t_0) tL \varepsilon_N^2 \log_N^3. \quad (4.6)$$

Because: A nonzero alias has total spatial momentum $\sigma \in \{\pm 2R_N, -4R_N\}$. Fix κ_1 and sum the other three factors by (3.4) at $\sigma - \kappa_1$, whose spatial modulus is at least R_N ; this is at most $C(tL)^2 \varepsilon_N^2 \log_N^2$. Sum the remaining envelope over the band by (3.1), giving $CtL \log_N$, and multiply by $4!(2Lt)^{-2}$.

⟨2⟩ 6 For the principal fourth-chaos difference, telescope the four products. Then

$$\|X_4 - Y_4\|_2^2 \leq C(m, t_0) t L \varepsilon_N^2 \log_N^3. \quad (4.7)$$

Because: Each telescoped common-mode term is bounded by

$$\sum_{\kappa \in D_N} |c_N - c_\infty|(\kappa) F_3(-\kappa).$$

Use (3.5), (3.4), sum k_0 in $(m^2 + k_0^2 + k^2)^{-3}$, and then sum $k^4 \log^2(2 + |k|)(m^2 + k^2)^{-5/2}$ up to R_N ; it is $O(tL \log_N^3)$. The common-mode result is $O((tL)^3 \varepsilon_N^2 \log_N^3)$ before the prefactor. Missing continuum modes use $c_{\text{env}} F_3$ and the $|k| \geq R_N$ tail, giving one fewer logarithm. Add (4.5), the one/two-edge bound, and (4.6).

⟨2⟩ 7 The second-chaos difference satisfies

$$\|X_2 - Y_2\|_2^2 = 2 \sum_{D_N} (c_N - c_\infty)^2 + 2 \sum_{D_\infty \setminus D_N} c_\infty^2 \leq C(m, t_0) t L \varepsilon_N^2. \quad (4.8)$$

Because: The two symmetric edge pairings have weights $1/2$ and restore full weight; the lattice half-open alias replaces the missing opposite edge, so no residual edge term remains. Square (3.5), sum k_0 with exponent 4, and note the remaining band sum is $O(LR_N^2)$, giving $O(tL \varepsilon_N^2)$. The continuum tail of the squared resolvent has the same order.

⟨2⟩ 8 Combining (2.2), (2.5), (4.7), and (4.8),

$$\|\mathcal{U}_N - \mathcal{U}\|_2 \leq C(m, \lambda, t_0)(1 + t)(1 + L)\varepsilon_N \log_N^{3/2}. \quad (4.9)$$

Because: Apply the triangle inequality to the fourth chaos, the $6dX_2$ term, the coefficient difference $(d_N - d_\infty)Y_2$, and the scalar difference $3(d_N^2 - d_\infty^2)2Lt$. Use $d_N, d_\infty \leq \bar{d}$ and $\|Y_2\|_2^2 \leq CtL$.

⟨2⟩ 9 This proves ⟨1⟩4. □

⟨1⟩ 5 Verify the Wick dictionary and absence of a continuum counterterm.

⟨2⟩ 1 Let

$$\delta_N = \frac{1}{4L} \sum_{k \in \Gamma_N} \left(\frac{1}{\gamma_N(k)} - \frac{1}{\gamma(k)} \right).$$

With $u_j = \pi j / (2r_N)$ and $\tilde{\delta} = m\varepsilon_N / 2$, pair the nonedge $\pm j$ terms to obtain

$$\delta_N = \frac{1}{2\pi} \sum_{j=1}^{r_N-1} g_{\tilde{\delta}}(u_j) \Delta u + E_{\text{edge}}, \quad g_{\tilde{\delta}}(u) = \frac{1}{\sqrt{\tilde{\delta}^2 + \sin^2 u}} - \frac{1}{\sqrt{\tilde{\delta}^2 + u^2}}, \quad (5.1)$$

where $0 \leq E_{\text{edge}} \leq \varepsilon_N / (8L)$.

Because: Substitute the two exact dispersions, use $\Delta u = \pi / (2r_N)$, and separate the unpaired $j = -r_N$ term.

⟨2⟩ 2 $|g'_{\tilde{\delta}}(u)| \leq C_0 < 6$ uniformly on $[0, \pi/2]$ and $\tilde{\delta} \geq 0$.

Because: Write $g_{\tilde{\delta}}(u) = \psi(\sin^2 u) - \psi(u^2)$ with $\psi(a) = (\delta^2 + a)^{-1/2}$. Use $u^2 - \sin^2 u \leq u^4/3$, $\sin u \geq 2u/\pi$, $|\sin 2u - 2u| \leq (2u)^3/6$, and $\psi''(a) = 3(\delta^2 + a)^{-5/2}/4$; the two derivative terms are bounded by $\frac{1}{2}(\pi/2)^5$ and $2/3$.

⟨2⟩ 3 The Riemann sum in (5.1) differs from its integral by $O(\varepsilon_N)$, and replacing $\tilde{\delta}$ by 0 changes the integral by $O(\varepsilon_N^2 \log(1/\varepsilon_N))$.

Because: The first claim is the Lipschitz Riemann bound. For the second, split the integral at $u = \tilde{\delta}$; the difference is at most Cu below the split and $C\tilde{\delta}^2/u$ above it.

⟨2⟩ 4 The limiting integral is

$$\frac{1}{2\pi} \int_0^{\pi/2} (\csc u - u^{-1}) du = \frac{1}{2\pi} \log \frac{4}{\pi}. \quad (5.2)$$

Because: The antiderivative difference is $\log \tan(u/2) - \log u$; its lower endpoint is $-\log 2$ and its upper endpoint is $-\log(\pi/2)$.

⟨2⟩ 5 Thus

$$\left| \delta_N - \frac{1}{2\pi} \log \frac{4}{\pi} \right| \leq C(m, L) \varepsilon_N. \quad (5.3)$$

⟨2⟩ 6 No quadratic or scalar counterterm survives in $\mathcal{U}$.

Because: At continuum truncation K , the field's thermal variance is $v_K = c_{L,K}^{\text{ct}} + d_K$. Recombining its own-variance Wick chaos by (2.1) with coefficient d_K gives exactly the vacuum-truncation Wick quartic. Since $d_K \rightarrow d_\infty$ exponentially, (2.5) is the $K \rightarrow \infty$ truncation-convention quartic. The finite offset (5.2) occurs simultaneously in the lattice field variance and in its Wick subtraction, so it cancels rather than becoming a mass term.

⟨2⟩ 7 This proves ⟨1⟩5. □

⟨1⟩ 6 Establish the shift-field estimates.

Fix a coarse level M , $N = M + r$, and a real coarse momentum symbol p_M . Define

$$h_N(s, x) = \frac{1}{2L} \sum_{k \in \Gamma_N} a_N(k, s) e^{ikx}, \quad a_N = \frac{\varepsilon_M^{1/2}}{2} P_r(\varepsilon_M k) \hat{p}_M(k) \frac{\sinh(\gamma_N(k)(s - t/2))}{\sinh(\gamma_N(k)t/2)}, \quad (6.1)$$

and h_∞ by replacing P_r, γ_N, Γ_N with $P_\infty, \gamma, \Gamma_\infty$.

⟨2⟩ 1 h_N, h_∞ are real and $\|h_N\|_\infty, \|h_\infty\|_\infty \leq \Sigma_M(p)$.

Because: Hermitian Fourier coefficients pair except at the half-open edge. For $r \geq 1$, $P_r(\pm\pi 2^r) = 0$ because the product contains $m_0(\pi) = 0$; for $r = 0$ the edge coefficient is real. The hyperbolic ratio in (6.1) has modulus at most one. Sum the D4 envelope $|P_r(\ell)| \leq C_D^{1/2} (1 + |\ell|)^{-1-\eta/2}$ by an integral comparison; the result is the explicit finite $\Sigma_M(p)$ of V2.

⟨2⟩ 2 For every integer $n \geq 1$,

$$\varepsilon_N \sum_x \int_0^t |h_N|^n \leq 2Lt \Sigma_M(p)^n, \quad \int_{\mathbb{T}_L} \int_0^t |h_\infty|^n \leq 2Lt \Sigma_M(p)^n. \quad (6.2)$$

Because: The spatial volumes are both $2L$; integrate the pointwise bound.

⟨2⟩ 3 With $\theta_H = (1 + \eta)/2$,

$$\sup_s \frac{1}{2L} \left[\sum_{k \in \Gamma_N} |a_N - a_\infty|^2 + \sum_{|k| \geq R_N} |a_\infty|^2 \right] \leq C_H 2^{-r\theta_H}. \quad (6.3)$$

Because: Split $a_N - a_\infty$ into filter and dispersion differences. On the band,

$$|P_r - P_\infty|(\ell) \leq C 2^{-r} |\ell| (1 + |\ell|)^{-1-\eta/2}.$$

Its squared sum is $O(2^{-r(1+\eta)})$. For the dispersion term use the γ derivative of the hyperbolic ratio, bounded by $t \coth(t_0 m/2)$, and $|\gamma_N - \gamma| \leq \varepsilon_N^2 k^4 / (24m)$. Split at $|\varepsilon_M k| = 2^{r/2}$: use this Taylor bound below and the trivial bound 2 above. Both pieces are $O(2^{-r(1+\eta)/2})$. The missing-mode D4 tail is $O(2^{-r(1+\eta)})$. The slowest exponent is θ_H .

⟨2⟩ 4 This proves ⟨1⟩6. □

⟨1⟩ 7 Prove the repaired normalized power-symbol estimates.

Let $B_N = \Gamma_N$, let $[k]_N$ be the representative modulo $2R_N$ in B_N , and, for $j = 1, 2, 3$, define

$$K_{N,j}^{\text{fold}}(p) = (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum k_{0,i}=p_0, [\sum k_i]_N=p}} \prod_i c_N(\kappa_i), \quad (7.1)$$

$$K_{\times,j}(\kappa) = (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum \kappa_i = \kappa}} \prod_i c_{\times}(\kappa_i), \quad (7.2)$$

$$K_{\infty,j}(\kappa) = (2Lt)^{-(j-1)} \sum_{\sum \kappa_i = \kappa} \prod_i c_{\infty}(\kappa_i). \quad (7.3)$$

⟨2⟩ 1 $\sup K_{a,j} \leq B_j(m, t_0)$ for all three families.

Because: For $j = 1$ use Step ⟨1⟩3⟨2⟩1. For $j = 2, 3$, dominate by F_2, F_3 in (3.3)–(3.4); the factors $(tL)^{j-1}$ cancel the normalization $(2Lt)^{j-1}$. Folded aliases are a finite sum of shifted exact outputs and satisfy the same supremum bound.

⟨2⟩ 2 On the fundamental band,

$$\sup_p |K_{N,j}^{\text{fold}}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2, \quad (7.4)$$

$$\sup_p |K_{\times,j}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2. \quad (7.5)$$

Because: Partition (7.1) into exact output and the finitely many nonzero shifts $2R_N n$. On exact common configurations telescope the j factors and use the flat consequence of (3.5), $|c_N - c_{\infty}| \leq C\varepsilon_N^2$, followed by the band sums of the remaining $j - 1$ envelope convolution; these are $O((tL)^{j-1} \log_N^{j-1})$. For a nonzero output shift, one convolution argument has spatial modulus at least R_N and (3.3)–(3.4) gives $O(\varepsilon_N^2 \log_N^{j-1})$ after normalization. For continuum configurations with one input outside the band, fix that input and use the same tail convolution; the result is $O(\varepsilon_N^2 \log_N)$. This proves (7.4). The proof of (7.5) is identical without folded output and using the second inequality in (3.5).

⟨2⟩ 3 For every exact output with $|k| \geq R_N$,

$$K_{\times,j}(k_0, k) + K_{\infty,j}(k_0, k) \leq C\varepsilon_N^2 \log_N^{j-1}. \quad (7.6)$$

Because: Apply (3.3) or (3.4) at spatial distance at least R_N and divide by $(2Lt)^{j-1}$.

⟨2⟩ 4 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the shifted-interaction comparison with no undefined remainder.

For $j = 1, 2, 3$, put $g_N = h_N^{4-j}$, $g_{\infty} = h_{\infty}^{4-j}$ and

$$T_{N,j} = \varepsilon_N \sum_x \int g_N : \eta_N^j :, \quad T_{\infty,j} = \int \int g_{\infty} : \eta^j :.$$

⟨2⟩ 1 The true mixed covariance-power contribution differs from the translation-invariant surrogate by at most

$$C(2Lt)^2 \Sigma_M(p)^{2(4-j)} \varepsilon_N \log_N^{j-1}. \quad (8.1)$$

Because: By (1.4), $\|\mathcal{R}_N\|_\infty \leq C\varepsilon_N$, while $\|C_\times\|_\infty \leq C \log_N$. Hence $|(C_\times + \mathcal{R}_N)^j - C_\times^j| \leq C\varepsilon_N \log_N^{j-1}$. Integrate against the pointwise bounds from (6.2).

⟨2⟩ 2 Write $b_N(p, k_0)$ and $b_\infty(k, k_0)$ for the spacetime Fourier coefficients of g_N, g_∞ . Extend b_N by zero outside B_N and set $\delta b = b_N - b_\infty$ on B_N , $\delta b = -b_\infty$ outside. Then the surrogate quadratic difference has the exact decomposition

$$\begin{aligned} Q_{N,j} - 2Q_{\times,j} + Q_{\infty,j} &= \langle \delta b, K_{\infty,j} \delta b \rangle + \langle b_N, (K_{N,j}^{\text{fold}} - K_{\infty,j}) b_N \rangle \\ &\quad - 2\langle b_N, (K_{\times,j} - K_{\infty,j}) b_{\infty, B_N} \rangle - 2A_{\times,j}, \end{aligned} \quad (8.2)$$

where

$$A_{\times,j} = \frac{1}{2Lt} \sum_{p \in B_N, n \neq 0} K_{\times,j}(p_0, p + 2R_N n) \overline{b_N(-p, -p_0)} b_\infty(p + 2R_N n, p_0). \quad (8.3)$$

Because: Expand all three Fourier pairings. The lattice–lattice spatial sums fold the exact output modulo $2R_N$, producing (7.1). In the mixed pairing, continuum integration fixes the exact output $p + 2R_N n$, while the lattice sum sees its representative p ; split $n = 0$ and $n \neq 0$. Add and subtract the continuum symbol on the fundamental band. Expanding $\langle \delta b, K_{\infty,j} \delta b \rangle$ gives the common-band continuum terms plus the entire continuum tail, leaving exactly (8.2)–(8.3). No term remains unnamed.

⟨2⟩ 3 The mixed alias-output term obeys

$$|A_{\times,j}| \leq C\varepsilon_N^2 \log_N^2 \|b_N\|_2 \|b_\infty\|_2. \quad (8.4)$$

Because: Only finitely many n occur: the support of a j -fold output lies in $[-jR_N, jR_N]$. For $n \neq 0$, (7.6) bounds the symbol. Apply Cauchy–Schwarz to the finite repeated coefficient sums; each coefficient occurs at most $2j + 1$ times.

⟨2⟩ 4 The coefficient norms satisfy

$$\|b_N\|_2^2 + \|b_\infty\|_2^2 \leq Ct \Sigma_M(p)^{2(4-j)}, \quad \|b_N - b_\infty\|_2^2 \leq Ct 2^{-r\theta_H}. \quad (8.5)$$

Because: Parseval and (6.2) give the first inequality. For the second, write the $(4 - j)$ -fold convolution difference as

$$a_N^{*n} - a_\infty^{*n} = \sum_{\ell=0}^{n-1} a_N^{*\ell} * (a_N - a_\infty) * a_\infty^{*(n-1-\ell)}.$$

Use $\ell^2 * \ell^1 \rightarrow \ell^2$, (6.3), and the uniform ℓ^1 bound from the D4 envelope. For a lattice alias, $|k_1 + \dots + k_n| \geq R_N$ forces one $|k_i| \geq R_N/n$; place that factor in the D4 ℓ^2 tail and the remaining factors in ℓ^1 . Missing continuum modes are identical. Integrate the uniform in s estimate over $[0, t]$.

⟨2⟩ 5 Equations (7.4)–(7.6), (8.2)–(8.5), and the Schur bound yield

$$\mathbb{E}[(T_{N,j} - T_{\infty,j})^2] \leq C \left[t 2^{-r\theta_H} + t \varepsilon_N^2 \log_N^2 + t^2 \varepsilon_N \log_N^{j-1} \right]. \quad (8.6)$$

Because: The first term of (8.2) is at most $B_j \|\delta b\|_2^2$. The next two use (7.4)–(7.5) times the coefficient norms. The last uses (8.4). Add the physical edge remainder (8.1), with the harmless volume factors absorbed in the fixed parameter constant.

⟨2⟩ 6 Since $\varepsilon_N = \varepsilon_M 2^{-r}$ and $\theta_H < 1$,

$$\varepsilon_N \log_N^2 \leq C(\varepsilon_M, \eta) 2^{-r\theta_H}. \quad (8.7)$$

Because: Divide by $2^{-r\theta_H}$; the ratio is $\varepsilon_M 2^{-r(1-\theta_H)}(1 + \log(1/\varepsilon_M) + r \log 2)^2$, whose supremum over $r \geq 0$ is finite because an exponential with positive exponent dominates a polynomial.

⟨2⟩ 7 The deterministic $j = 0$ term has absolute difference $O(t 2^{-r\theta_H/2})$.

Because: It is the zero Fourier coefficient of $h_N^4 - h_\infty^4$. Use the same four-factor convolution telescope and alias/missing-mode estimate as in (8.5), followed by Cauchy–Schwarz.

⟨2⟩ 8 After Wick-binomial coefficients and the $d_N - d_\infty$ recombination errors are included,

$$\|[\mathcal{U}_N(\eta_N + h_N) - \mathcal{U}_N(\eta_N)] - [\mathcal{U}(\eta + h_\infty) - \mathcal{U}(\eta)]\|_2 \leq C(1+t)^{5/2} \left[2^{-r(1+\eta)/4} + \varepsilon_N \log_N \right]. \quad (8.8)$$

Because: Take square roots in (8.6), use $\theta_H/2 = (1+\eta)/4$, sum $j = 0, 1, 2, 3$ with coefficients 1, 4, 6, 4, and use $|d_N - d_\infty| = O(\varepsilon_N^2)$. The largest volume power is bounded by $(1+t)^{5/2}$.

⟨2⟩ 9 This proves ⟨1⟩8. □

⟨1⟩ 9 Prove the uniform Nelson exponential bounds.

⟨2⟩ 1 Pointwise, for real u ,

$$:u^4:_c = (u^2 - 3c)^2 - 6c^2 \geq -6c^2.$$

Hence

$$\mathcal{U}_N(\eta_N + h) \geq -c_1 t \log_N^2 \quad (9.1)$$

for every deterministic real shift h .

Because: Integrate the pointwise inequality over volume $2Lt$ and use $c_{N,L} \leq C(m, L) \log_N$. Replacing u by the real number $u + h$ changes nothing.

⟨2⟩ 2 For $b \leq c_1 t \log_N^2$, choose the largest dyadic level $N_b \leq N$ with $1 + \log(1/\varepsilon_{N_b}) \leq L_b = (b/(2c_1 t))^{1/2}$. Then

$$\{\mathcal{U}_N \leq -b\} \subseteq \{\mathcal{U}_N - \mathcal{U}_{N_b} \leq -b/2\}. \quad (9.2)$$

Because: Dyadic log increments are exactly $\log 2$, so for sufficiently large b the level exists and its logarithm is at least $L_b/2$. By (9.1), $\mathcal{U}_{N_b} \geq -c_1 t L_b^2 = -b/2$.

⟨2⟩ 3 The comparison estimates imply

$$\rho_b := \|\mathcal{U}_N - \mathcal{U}_{N_b}\|_2 \leq C L_b^{3/2} e^{-c L_b}. \quad (9.3)$$

Because: Apply (4.9) at levels N, N_b through the common continuum interaction; $\varepsilon_{N_b} \leq e^{1-L_b/2}$. In the shifted case also use (8.8); its dyadic term is $O(e^{-(1+\eta)L_b/8})$. Take the smaller positive exponent c .

⟨2⟩ 4 If Z is a sum of Gaussian chaoses of order at most four, then for $q \geq 1$,

$$\|Z\|_{2q} \leq (2q-1)^2 \|Z\|_2. \quad (9.4)$$

Because: Write $Z = \sum_{j=0}^4 Z_j$ orthogonally. For $\tau = \frac{1}{2} \log(2q-1)$, set $Y = \sum e^{j\tau} Z_j$; then $Z = \Gamma(e^{-\tau})Y$. Nelson hypercontractivity gives $\|Z\|_{2q} \leq \|Y\|_2 \leq e^{4\tau} \|Z\|_2 = (2q-1)^2 \|Z\|_2$.

⟨2⟩ 5 Chebyshev with $(2q)^2 = e^{cL_b/2}$ gives

$$\mathbb{P}(\mathcal{U}_N \leq -b) \leq \exp[-e^{cL_b/4}] \quad (9.5)$$

for all sufficiently large b , uniformly in N ; the same holds for shifted interactions on bounded coarse-symbol sets.

Because: By (9.2)–(9.4),

$$\mathbb{P}(\mathcal{U}_N \leq -b) \leq \left[\frac{2}{b} (2q)^2 \rho_b \right]^{2q}.$$

Insert (9.3); for large b the bracket is at most e^{-1} , and $2q = e^{cL_b/4}$.

⟨2⟩ 6 For every $p < \infty$ and $t \in [t_0, T]$,

$$\sup_N \mathbb{E}[e^{-p\mathcal{U}_N}] < \infty, \quad \sup_N \mathbb{E}[e^{-p\mathcal{U}_N(\eta_N + h_N)}] < \infty. \quad (9.6)$$

Because: The layer-cake formula integrates $pe^{pb}\mathbb{P}(\mathcal{U}_N \leq -b)$. The double-exponential tail (9.5) dominates e^{pb} ; the bounded b interval contributes a finite uniform constant.

⟨2⟩ 7 This proves ⟨1⟩9. □

3 V2 conclusion

The repaired V2 outputs are (4.9), the Wick dictionary (5.3) with no residual counterterm, the shift-field convergence (6.3), the fully accounted shifted comparison (8.8), and the unshifted/shifted Nelson bounds (9.6). The original four-edge assertion and the original R_{tail} line must not be used; equations (4.4)–(4.5) and (8.2)–(8.4) replace them.